

Curriculum Vitae

Nora Morris

March 26, 2026

COURSEWORK

(Hendrix College)

Linear Algebra - Fall 2020
Introduction to Advanced Mathematics - Spring 2021
Multivariable Calculus - Spring 2021
Abstract Algebra - Fall 2021
Mathematical Models - Fall 2021
Probability and Statistics - Spring 2022
Scientific Computing - Fall 2022
Intermediate Chinese - Fall 2022
Combinatorics - Fall 2022
Programming Languages - Spring 2023
Interactive Game Development - Spring 2023
Complex Variables - Spring 2023
Artificial Intelligence - Fall 2023
Algorithms - Fall 2023
Computing Systems Organization - Spring 2024
Functional Programming - Spring 2024

(University of Illinois Chicago)

Introduction to Cryptography - Fall 2024
Applied Artificial Intelligence - Fall 2024
Economics and Computation - Spring 2025
Algorithms II - Spring 2025
Operating Systems and Concurrency - Fall 2025
Advanced Topics in Economics and Computation - Fall 2025

EMPLOYMENT

Software Developer @ Disco Tray Studios - Conway, AR - Aug 2023 - PRESENT Using C# and Unity to implement software for local clients. Collaborating with other developers using Github and GitKraken.

Calculus Tutor @ Math Help Center — Conway, AR — Sep 2021 - May 2023 - Assisted Hendrix students with Precal through Calc II in person

Research Assistant @ UIC — Chicago, IL — June 2025 - August 2025
- Conducted below research project

Teacher's Assistant @ UIC — Chicago, IL — August 2024 - PRESENT
- Graded and Assisted Students in Discrete Math, Advanced Data Structures, & Economics and Computation.

RESEARCH PROJECTS

Combinatorics of Square Grid Regions

Devised original combinatoric techniques to enumerate partitions of square grids of variable side length, and wrote a version of Simon Tatham's *Galaxies* using python and LaTeX. Presented publicly at the Hendrix-Rhodes-Sewanee mathematics conference in 2023.

Representing Incomplete Information in Game Description Language and Monte Carlo Tree Search

Wrote the first card games for open-source General Game System Ludii, and wrote an AI system capable of intelligent, non-clairvoyant play of general card games. Presented publicly at Hendrix in 2023.

Abstract Cellular Automaton Builder

Created a Haskell schema for simulating arbitrary cellular automata, with parameters for neighborhood, number of states, the transition matrix, and more.

The Tokenizer

Attempted four methods to translate English to a modular 150-word constructed language (Toki Pona), in the interest of semantic compression and automated understanding.

OpenSpiel Research Project - Implemented several general algorithms for finding nash equilibria in imperfect information games using openspiel, including novel variants and research.